

Topology MATM36: Course Summary of Important Results

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Metric Spaces

Definition 1. Let X be a topological space and $A \subseteq X$. A point $x \in X$ is called an adherent point of A if every open neighbourhood U of x satisfies

$$U \cap A \neq \emptyset.$$

In the special case where X is a metric space with metric d , this is equivalent to requiring that for every $r > 0$,

$$B_r(x) \cap A \neq \emptyset.$$

Definition 2. Let X be a topological space and $A \subseteq X$. The set A is called dense in X if every nonempty open set $U \subseteq X$ satisfies

$$U \cap A \neq \emptyset.$$

Equivalently, A is dense in X if $\overline{A} = X$.

Definition 3. Let X be a topological space and $A \subseteq X$. The set A is called nowhere dense in X if

$$\text{int}(\overline{A}) = \emptyset.$$

Definition 4. A topological space X is separable if: $\exists D \subseteq X : D$ countable and dense.

Baire Category Theorem

Theorem 1 (Baire Category Theorem). Let (X, d) be a complete metric space and let $\{U_n\}_{n=1}^\infty$ be a sequence of dense open subsets of X . Then $\bigcap_{n=1}^\infty U_n$ is dense in X .

Proof. Fix a nonempty open ball $B(x, \varepsilon) \subset X$.

Since U_1 is dense and open, choose $y_1 \in U_1 \cap B(x, \varepsilon)$ and $r_1 > 0$ such that

$$\overline{B}(y_1, r_1) \subset U_1 \cap B(x, \varepsilon), \quad r_1 < 1.$$

Inductively, having chosen y_n, r_n , use that U_{n+1} is dense and open to pick $y_{n+1} \in U_{n+1} \cap B(y_n, r_n)$ and $r_{n+1} > 0$ such that

$$\overline{B}(y_{n+1}, r_{n+1}) \subset U_{n+1} \cap B(y_n, r_n), \quad r_{n+1} < \frac{1}{n+1}.$$

Then the closed balls are nested:

$$\overline{B}(y_{n+1}, r_{n+1}) \subset \overline{B}(y_n, r_n) \subset \cdots \subset \overline{B}(y_1, r_1) \subset B(x, \varepsilon),$$

and for $m > n$ we have $y_m \in B(y_n, r_n)$, hence $d(y_m, y_n) < r_n \rightarrow 0$. Thus (y_n) is Cauchy, so by completeness $y_n \rightarrow y \in X$.

For each fixed n , all tails of the sequence lie in $\overline{B}(y_n, r_n)$, so the limit satisfies $y \in \overline{B}(y_n, r_n) \subset U_n$. Hence $y \in \bigcap_{n=1}^{\infty} U_n$, and also $y \in \overline{B}(y_1, r_1) \subset B(x, \varepsilon)$. Therefore

$$B(x, \varepsilon) \cap \bigcap_{n=1}^{\infty} U_n \neq \emptyset,$$

so the intersection is dense. □

Remark 1. From the definitions it follows that a set $B \subseteq X$ satisfies $\text{int}(B) = \emptyset \iff X \setminus B$ is dense in X . In particular, if A is nowhere dense, then $X \setminus \overline{A}$ is open and dense in X .

Corollary 1. Let (X, d) be a complete metric space and let $\{F_n\}_{n=1}^{\infty}$ be a sequence of nowhere dense subsets of X . Then

$$\text{int}\left(\bigcup_{n=1}^{\infty} F_n\right) = \emptyset.$$

Proof. Since F_n is nowhere dense, $U_n := X \setminus \overline{F_n}$ is open and dense; by Baire, $\bigcap_{n=1}^{\infty} U_n = \bigcup_{n=1}^{\infty} \overline{F_n}$ is dense, hence nonempty in every nonempty open set. □

Finite Products of Metric Spaces

Definition 5. Let (X_k, d_k) , $k = 1, \dots, n$, be metric spaces and set

$$X = X_1 \times \cdots \times X_n.$$

A metric d on X is said to be a product metric if convergence in (X, d) is coordinatewise,

$$x^{(m)} = (x_1^{(m)}, \dots, x_n^{(m)}) \rightarrow x = (x_1, \dots, x_n)$$

in X if and only if for each $k = 1, \dots, n$,

$$x_k^{(m)} \rightarrow x_k \quad \text{in } X_k.$$

Theorem 2. Suppose d is a metric on $X = X_1 \times \cdots \times X_n$ satisfying the above coordinatewise convergence property. Then the open sets in (X, d) are precisely the unions of product sets

$$U_1 \times \cdots \times U_n,$$

where each U_k is open in X_k .

Proof. First, if $E_k \subset X_k$ is closed for each k , then

$$E_1 \times \cdots \times E_n$$

is closed in X . Indeed, if $x^{(m)} \in E_1 \times \cdots \times E_n$ and $x^{(m)} \rightarrow x$ in X , then by coordinatewise convergence, $x_k^{(m)} \rightarrow x_k$ in X_k . Since each E_k is closed, $x_k \in E_k$, hence $x \in E_1 \times \cdots \times E_n$.

Now let $U_k \subset X_k$ be open. Then

$$V^{(k)} := X_1 \times \cdots \times X_{k-1} \times U_k \times X_{k+1} \times \cdots \times X_n$$

is open in X , since its complement is of the form

$$X_1 \times \cdots \times (X_k \setminus U_k) \times \cdots \times X_n,$$

which is closed by the previous remark. Hence

$$U_1 \times \cdots \times U_n = \bigcap_{k=1}^n V^{(k)}$$

is open in X , and any union of such products is open.

Conversely, let $U \subset X$ be open and $x = (x_1, \dots, x_n) \in U$. We show there exist open sets $U_k \subset X_k$ such that

$$x \in U_1 \times \cdots \times U_n \subset U.$$

Suppose not. Then for each $m \in \mathbb{N}$,

$$B_{X_1}(x_1, 1/m) \times \cdots \times B_{X_n}(x_n, 1/m) \not\subset U.$$

Thus there exists $x^{(m)} \notin U$ with

$$d_k(x_k^{(m)}, x_k) < \frac{1}{m}, \quad k = 1, \dots, n.$$

Then $x_k^{(m)} \rightarrow x_k$ for each k , hence $x^{(m)} \rightarrow x$ in X . Since $X \setminus U$ is closed and each $x^{(m)} \in X \setminus U$, we conclude $x \in X \setminus U$, a contradiction. Therefore such open sets U_k exist, and U is a union of product sets. $\square$